

24k-0569

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BS-03-1H.

01) (a). $\lim_{x \rightarrow 0} \left(\frac{1}{x \sin x} - \frac{1}{x \tan x} \right) \rightarrow \lim_{x \rightarrow 0} \frac{1}{x \sin x} - \frac{1}{\frac{x \sin x}{\cos x}} \rightarrow \frac{1 - \cos x}{x \sin x} \frac{d/dx}{d/dx}$ $u = x$
 $\lim_{x \rightarrow 0} \frac{\sin x}{x \cos x + \sin x} \frac{d/dx}{d/dx} \leftarrow = \text{if } \lim_{x \rightarrow 0} \frac{\sin x}{(x \cos x) + (\sin x)} \rightarrow \frac{0}{0} \text{ [again]}$
 $\lim_{x \rightarrow 0} \frac{\cos x}{(-x \sin x + \cos x) + \cos x} \rightarrow \lim_{x \rightarrow 0} \frac{\cos(0)}{(-0 + (0 + x \cos(0)) + \cos(0))} \rightarrow \boxed{\frac{1}{2}}$

(b). $\lim_{x \rightarrow -\infty} (x - \sqrt{x^2 + 3x})$

$\lim_{x \rightarrow -\infty} \frac{(x - \sqrt{x^2 + 3x})(x + \sqrt{x^2 + 3x})}{(x + \sqrt{x^2 + 3x})} = \frac{x^2 - x^2 - 3x}{(x + \sqrt{x^2 + 3x})} \rightarrow \frac{-3x}{(x + \sqrt{x^2 + 3x})}$
 $\lim_{x \rightarrow -\infty} = \frac{-3x}{x + \sqrt{x^2(1 + 3/x)}} \rightarrow \frac{-3x}{x + x\sqrt{1 + 3/x}} = \frac{-3}{1 + 1} = \frac{-3}{1 + \sqrt{1 + 3/x}} \rightarrow \frac{-3}{2}$

(c). $\lim_{x \rightarrow +\infty} (x - \ln(x^2 + 3))$

$\lim_{x \rightarrow +\infty} \frac{(x - \ln(x^2 + 3))(x + \ln(x^2 + 3))}{x + \ln(x^2 + 3)}$
 $\lim_{x \rightarrow +\infty} \frac{x^2 - [\ln(x^2 + 3)]^2}{x + \ln(x^2 + 3)} \xrightarrow{\frac{\infty}{\infty} \rightarrow \frac{d/dx}{d/dx}} \frac{2x - 2[\ln(x^2 + 3)] \cdot x \cdot \frac{2x}{x^2 + 3}}{1 + \frac{1}{x^2 + 3} (2x)}$
 $\lim_{x \rightarrow +\infty} = \frac{2x - 4x \frac{[\ln(x^2 + 3)]}{x^2 + 3}}{1 + \frac{2x}{x^2 + 3}}$ again $\frac{d/dx}{d/dx}$

MAAZ

Q2). $f(x) = 5x^{1/3} - x^{5/3}$

$$f'(x) = 0$$

$$0 = \frac{1}{3} (5x^{1/3-1}) - \frac{5}{3} x^{5/3-1}$$

$$0 = \frac{5}{3x^{2/3}} - \frac{5x^{2/3}}{3}$$

$$0 = \frac{5}{3x^{2/3}} - \frac{5x^{2/3}}{3}$$

$$0 = \frac{5 - 5x^{4/3}}{3x^{2/3}}$$

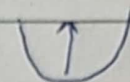
$$0 = 5 - 5x^{4/3}$$

$$x^{4/3} = 1 \rightarrow \text{Take cube on both sides}$$

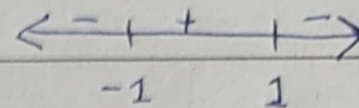
$$(x^{4/3})^3 = (1)^3 \rightarrow x^4 = 1$$

$$x = \pm 1$$

local minima at $x = -1$



local maxima at $x = 1$



$$\text{At } x = -2 \rightarrow (-)$$

$$\text{At } x = 0 \rightarrow (+)$$

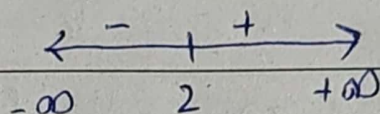
$$\text{At } x = 2 \rightarrow (-)$$

Q3). $f(x) = x^3 - 6x^2 + 9x + 30$

$$f'(x) = 3x^2 - 12x + 9$$

$$f''(x) = 6x - 12 = 0$$

$$x = 2$$



$$\text{at } x = 1 \rightarrow -6 \text{ so } (-)$$

$$\text{at } x = 3 \rightarrow 6 \text{ so } (+)$$

concave up $\rightarrow (2, +\infty)$

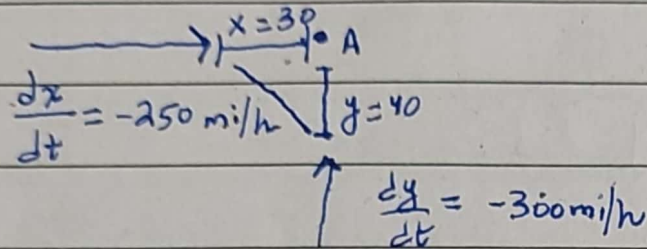
So inflection points at $x = 2$

concave down $\rightarrow (-\infty, 2)$

$$\text{so } f(2) = 2^3 - 6(2)^2 + 9(2) + 30 \rightarrow 32 \rightarrow y$$

so inflection point $(2, 32)$.

Q4). Ans:-



$$x = 30 \quad y = 40 \quad z = 50$$

$$\frac{dx}{dt} = -250 \text{ mi/h} \quad \frac{dy}{dt} = -300 \text{ mi/h} \quad \frac{dz}{dt} = ?$$

$$z = \sqrt{x^2 + y^2} \rightarrow z = \sqrt{30^2 + 40^2} \rightarrow (50)$$

$$z^2 = x^2 + y^2 \rightarrow \text{Differentiate it}$$

$$2z \frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 2(50) \frac{dz}{dt} = 2(30)(-250) + 2(40)(-300)$$

$$\boxed{\frac{dz}{dt} = -390 \text{ mi/h}}$$

Q5). Ans:- $f(x) = \sqrt{9 - (x-3)^2}$ $n=4$, L_4 , R_4

$$[0, 6]$$

$$\Delta x = \frac{b-a}{n} = \frac{6-0}{4} = \frac{6}{4} = 1.5$$

So for $L_4 [0, 6] \rightarrow 0, 1.5, 3, 4.5 \rightarrow \Delta x \times \sum_{n=1}^4 f(x_n) \rightarrow 1.5 [f(0) + f(1.5) + f(3) + f(4.5)]$

for $R_4 [0, 6] \rightarrow 6, 4.5, 3, 1.5 \rightarrow$

$$R_4 = 1.5 [f(6) + f(4.5) + f(3) + f(1.5)]$$

$$= 1.5 (0 + 2.598 + 3 + 2.598)$$

$$\boxed{L_4 = 12.294}$$

$$R_4 = 1.5 [0 + 2.598 + 3 + 2.598] \rightarrow R_4 = 12.294$$

$$\boxed{R_4 = 12.294}$$

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